Review and assessment of measured values of the nonlinear refractive-index coefficient of fused silica

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The literature describes more than 30 measurements, at wavelengths between 249 and 1550 nm, of the absolute value of the nonlinear refractive-index coefficient of fused silica. Results of these experiments were assessed and best currently available values were selected for the wavelengths of 351, 527, and 1053 nm. The best values are $(3.6 \pm 0.64) \times 10^{-16}$ cm^2/W at 351 nm, $(3.0 \pm 0.35) \times 10^{-16}$ cm^2/W at 527 nm, and $(2.74 \pm 0.17) \times 10^{-16}$ cm^2/W at 1053 nm. © 1998 Optical Society of America

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1. Introduction

Fused silica is currently the material of choice for fabrication of lenses and windows for lasers that operate at high intensity and at wavelengths from 250 to 1064 nm. It is highly transparent, largely free of inclusions, and mechanically stable. The intensity that can be tolerated by a single thin silica component is limited by laser-induced damage. In a thin element, damage usually occurs at a polished surface or in an optical coating. As the length of the beam path through silica increases, small-scale self-focusing becomes important.[1] This self-focusing occurs through the amplification of amplitude noise on the beam, which can be thought of as weak parasitic beams that propagate at an angle to the primary beam. In a fusion laser, small-scale self-focusing increases the depth of intensity ripple on a beam, causes overloading of the pinholes in spatial filters, contributes to reduction of the efficiency of harmonic conversion, and leads to generation of light that will miss the fusion target. Therefore, the self-focusing coefficient and the threshold for laser-induced damage are two key design parameters for a laser.

Self-focusing is caused by an intensity-dependent change in the refractive index. When the laser pulse is of short duration, and the glass is an isotropic material such as silica, the induced change in index can be specified by a single parameter γ. The defining relationship is $n = n_0 + \gamma I$. Here n_0 is the linear refractive index and I is the intensity.

Recently it became necessary to define values of γ for use in the propagation code PROP2[2] that is being used to analyze proposed designs of the laser for the National Ignition Facility (NIF).[3] The laser will employ amplifiers made of Nd-doped phosphate glass and operate at a wavelength of 1053 nm. Fused silica will be used to fabricate passive components such as lenses in the amplifier chains. Immediately before a beam enters the target chamber, it will pass through a frequency doubler and a mixer, both fabricated of potassium dihydrogen phosphate (KDP). A large fraction of the energy will be converted to the third harmonic at 351 nm. The unconverted residual usually will consist primarily of 1053-nm light. The intense 351-nm light will then pass into the target chamber through the final silica components, which are a focusing lens, a debris shield, and possibly a grating etched into a silica window. Therefore modeling of the laser requires values of γ for both silica and KDP at all three harmonic wavelengths and a 1053-nm value for the Nd:glass.

Currently one must rely on measured values of γ because an accurate technique for calculating this coefficient has not been developed. The measurements are quite difficult. Over a span of 28 years, more than 30 absolute measurements of γ in silica have been made at wavelengths ranging from 249 to 1550 nm. At some wavelengths, the measured values differ by a factor of more than 3. A review of these measurements is provided in Section 2. Section 3 is a brief discussion of dispersion of γ. Criteria that were used to select values of γ for use in design of the NIF are described in Section 4.

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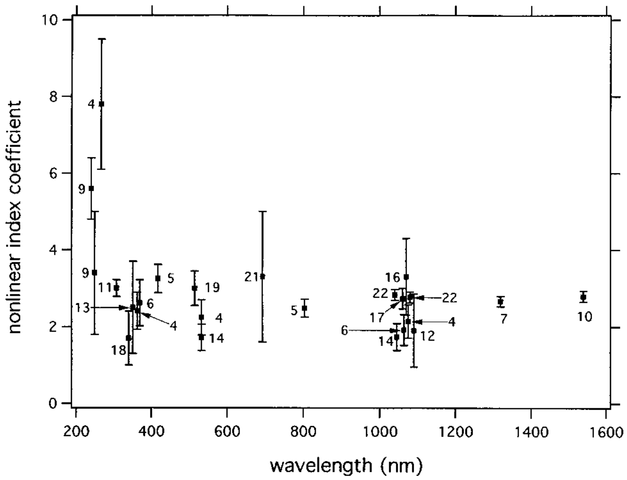

Fig. 1. Measured values of the nonlinear refractive-index coefficient as a function of wavelength. Coefficients are plotted in multiples of 1×10^{-16} cm^2/W. To make the error bars legible, some data were plotted at wavelengths slightly removed from the measurement wavelength. The source for each datum is indicated by the attached reference number.

Table 1. Measured Values of the Nonlinear Refractive-Index Coefficient γ for Silica[a]

Nonlinear Coefficient	Wavelength (nm)	Reference Number
5.6 ± 0.8	248	9
3.4 ± 1.6	248	9
7.8 ± 0.17	266	4
3.0 ± 0.22	308	11
1.7 ± 0.7	351	18
2.5 ± 1.2	351	13
2.41 ± 0.48	355	4
2.62 ± 0.6	355	6
3.42 ± 0.37	402	5
3.0 ± 0.45	514	19
1.72 ± 0.34	532	14
2.24 ± 0.46	532	4
3.3 ± 1.7	694	21
2.48 ± 0.23	804	5
2.77 ± 0.14	1053	22
2.83 ± 0.14	1053	22
1.73 ± 0.35	532	14
1.9 ± 0.95	1064	12
2.73 ± 0.27	1064	17
1.92 ± 0.4	1064	6
3.3 ± 1.0	1064	16
2.14 ± 0.43	1064	4
2.66 ± 0.13	1319	7
2.79 ± 0.14	1550	10

[a]Values are in multiples of 1×10^{-16} cm^2/W.

2. Literature Citations and Initial Sorting of the Data

The papers that describe absolute measurements of γ for silica are cited, in reverse chronological order, in Refs. 4–22. A measurement was deemed to be absolute, and the resulting data were included in the database if

(1) the experimental technique allowed measurement of all the parameters that were involved in the calculation of γ;
(2) an uncertainty was assigned, allowing the datum to be meaningfully compared with results from other experiments; and
(3) the measurement had not been superseded by a later measurement by the same authors.

Values of γ that were generated in an experiment that met these general criteria were taken at face value, except for a single case. Three data from one paper[8] were omitted on the recommendation of the authors (comment within the paper) who expressed concern about the accuracy of the Raman standard for their experiment.

Figure 1 shows the data that were selected by this initial sorting and the associated uncertainties. To allow legibility for all the uncertainty ranges, some data were plotted at wavelengths that are slightly removed from the actual wavelength of the experiment. The source of each datum is given by the attached reference number, and the numerical values of γ are given in Table 1. This initial sorting eliminated a few of the values that have been propagated through general reviews of self-focusing coefficients.[23–25]

3. Dispersion of γ

Two models of the dispersion of γ have been described in the literature. One model, the PERT equation, is based on a perturbation theory[8] that assumes that dispersion of γ in the visible and near IR can be attributed to a single resonance in the UV. The position of that resonance has been calculated for silica.[8] The other model is based on a Kramers–Kronig transformation.[26] It provided a general description of the dispersion in measured values of γ in semiconductors, but it did not agree as closely with data for wide-bandgap dielectrics.[4] Both models indicate that γ is larger at 351 nm than at 1053 nm (normal dispersion), but the models do not agree in detail as to the shape of the dispersion curve, and neither provides accurate absolute values. However, the PERT equation is the logical choice for the present application because it more nearly describes the dispersion of measured γ values for silica.

4. Selection of Best Values at Wavelengths of 351, 527, and 1053 nm

The task of selecting the best values of γ was complicated greatly by the spread in the measured values. A simple inspection of the data, for any narrow range of wavelengths, provides three possibilities. Either some of the data are incorrect, some of the uncertainties were under estimated, or mechanisms other than electronic self-focusing were observed during some of the experiments. The third of these possibilities might have applied to one of the experiments.[10] It employed a cw pump beam that was modulated at 7

Table 2. Eleven Values of the Nonlinear Refractive-Index Coefficient[a]

Nonlinear Coefficient	Wavelength (nm)	Reference Number
2.40 ± 0.22	804	5
2.77 ± 0.14	1053	22
2.83 ± 0.14	1053	22
1.73 ± 0.35	1064	14
1.9 ± 0.95	1064	12
2.73 ± 0.27	1064	17
1.92 ± 0.4	1064	6
3.3 ± 1.0	1064	16
2.14 ± 0.43	1064	16
2.71 ± 0.13	1319	7
2.87 ± 0.14	1550	10

[a]Values used in the first of the error-bar weighted averages at 1053 nm.

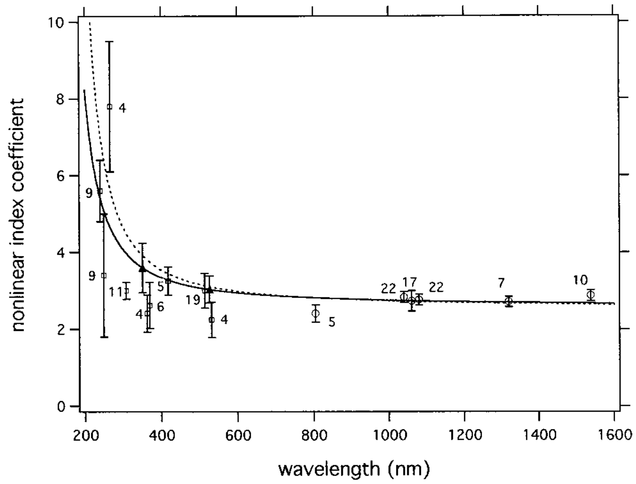

Fig. 2. Values of γ that remained after the selection process and designated best values at 351 and 527 nm. Coefficients are plotted in multiples of 1×10^{-16} cm^2/W. The solid and dashed curves are PERT curves with UV resonances at, respectively, 1.45×10^5 cm^{-1} and 1.25×10^5 cm^{-1}.

Mhz, and there is some possibility that the observed phase shift was partially caused by electrostriction. Even if that were true, the consequence is small because reevaluation of only one of 24 data cannot have a significant impact on the conclusions that are drawn from that database.

To allow selection of best values from this database, the following somewhat arbitrary process was designed. First, an error-bar weighted averaging was used to select the 1053-nm value. Then a modified PERT function was scaled in amplitude so that it passed through the selected 1053-nm value, and values for the wavelengths of 527 and 351 nm were read from the dispersion curve. No effort was made to define a value for the fourth harmonic at 263 nm.

The decision to start with selection of the 1053-nm value was made because it allowed us to average 11 data. Eight experiments, some of them among the most precise, were performed at either 1053 or 1064 nm. Three additional measurements were made at wavelengths of 804, 1319, and 1550 nm. Both models agree that dispersion is weak over this wavelength range. The PERT equation predicts that γ decreases by approximately 5% between 804 and 1550 nm, and it cannot distinguish values of γ for 1053 and 1064 nm. This equation was used to translate the three outlying data to 1064 nm. The required scalings were -3% for the 804-nm datum and $+2\%$ for the 1550-nm datum. The assemblage of eleven 1064-nm values is given in Table 2.

The best 1064-nm value was determined from a weighted average of the γ values in Table 2. The weight of each datum was inversely proportional to its uncertainty. The uncertainty of the average was taken to be the root mean square of the individual uncertainties. With all 11 data included, the average was $\gamma = (2.56 \pm 0.3) \times 10^{-16}$ cm^2/W. Data that fell outside this uncertainty range were deleted, and the average was recalculated. The result was $\gamma = (2.74 \pm 0.17) \times 10^{-16}$ cm^2/W. This second value agrees with all the most precise data at 1064 nm and has been selected for use in the design in the NIF.

Weighted averaging was not a useful technique for selecting best values for 351 and 527 nm. Within this range, there are fewer precise data and none with the precision that is claimed for some of the recent IR data. Also, it is suggested by the general trends in the data, and by both dispersion theories, that dispersion is stronger in the UV. It is less likely that this dispersion is modeled accurately by either of the theories.

The process for selecting values at 351 and 527 nm was initiated by the deletion of four data. Two early experiments yielded results (2.5 ± 1.2 at 351 nm and 3.3 ± 1.7 at 694 nm) with uncertainties so large that these data were of little assistance in refining the value of γ. Two other data, 1.7 ± 0.7 at 351 nm and 1.72 ± 0.34 at 351 nm, when compared with the selected 1053-nm value, indicate negative dispersion for γ for the wavelength range from 1053 to 351 nm. These four data were deleted. The remainder of the visible and UV data, the IR values that survived the error-bar weighted averaging, and two PERT functions are plotted in Fig. 2.

The dashed curve is the PERT function for silica. Its resonance is at 1.25×10^5 cm^{-1},[8] and its amplitude was scaled to agree with the averaged IR value 2.74×10^{-16} cm^2/W. As expected, this scaling provides excellent agreement with all the precise IR data. The curve agrees with two of the data at 402 and 514 nm, but it predicts values larger than several of the UV data.

If the shape of the PERT function is retained, the only option for improving the agreement of this function with the UV data is to rescale its amplitude. One could argue, in fact, that the function should be scaled to agree with the result of the initial averaging of the eleven 1053-nm values, 2.56×10^{-16} cm^2/W. This option was rejected because it provided only a small improvement in the UV, and it destroyed the agreement with the precise IR data.

A second option is to distort slightly the shape of the PERT function for silica by repositioning its resonance. The solid curve in Fig. 2 is a PERT function with resonance at 1.45×10^5 cm^{-1}. This curve is a better fit to the UV data, and it agrees with the precise IR data. Values of γ that are currently being used in the design of the NIF were read from this curve; $\gamma = (3.0 \pm 0.35) \times 10^{-16}$ cm^2/W at 527 nm and $\gamma = (3.6 \pm 0.64) \times 10^{-16}$ cm^2/W at 351 nm. The uncertainties were selected arbitrarily to be large enough to account for the majority of the data at nearby wavelengths.

5. Summary

Measurements of the nonlinear refractive-index coefficient of silica were reviewed. We used minimal criteria for defining an absolute measurement to determine which of the reported values should be included in the data set, and a comprehensive list of measured values was prepared. In most cases, application of these criteria was straightforward, so the table of data itself should not be controversial.

The best values for three wavelengths were selected by a procedure that was somewhat arbitrary. A 1053-nm value was selected by error-bar weighted averaging, and this allowed domination of the average by a few precise, recently published data. The remainder of the data contributed primarily to the calculated uncertainty of the average.

These precise data, coupled with an insistence that the dispersion was normal, also strongly influenced the choice of values for the wavelengths of 351 and 527 nm. Values for these wavelengths were read from a dispersion curve that had been slightly distorted, and uncertainties large enough to account for most of the data were assigned.

There is room for improvement in our knowledge of γ in silica, although it is the author's opinion that the uncertainty is not nearly so large as indicated in a recent publication.[8] And, although it is important to obtain a general understanding of the dispersion γ, values at 351 and 1053 nm are of greatest interest for designers of the NIF. Experimental studies of small-scale self-focusing have been done at both these wavelengths to provide data for comparison with the NIF design codes. Initial modeling of these experiments by the code PROP2 indicates that the majority of the results can be explained with the values of γ that were selected by this review of the literature. Details of these experiments will be published upon completion of the data reduction.

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